

MooreVIEW Lift Station — Parc & Cloud Build Guide

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Audience: Panel builders, field technicians, cloud commissioning engineers
I/O reference: [LIFT-STATION-OPTA-IO-MAP.md](#) · PDF: [MooreVIEW-Lift-Station-Opta-IO-Map.pdf](#)

Purpose

Single build and commissioning guide for **simplex**, **duplex**, and **triplex** lift-station control panels on **Arduino Opta + MQTT Parc + Cloud Studio**.

Covers panel wiring, firmware flash, Opta `/setup` , cloud driver deploy, and hardware bench testing.

1. Configuration matrix

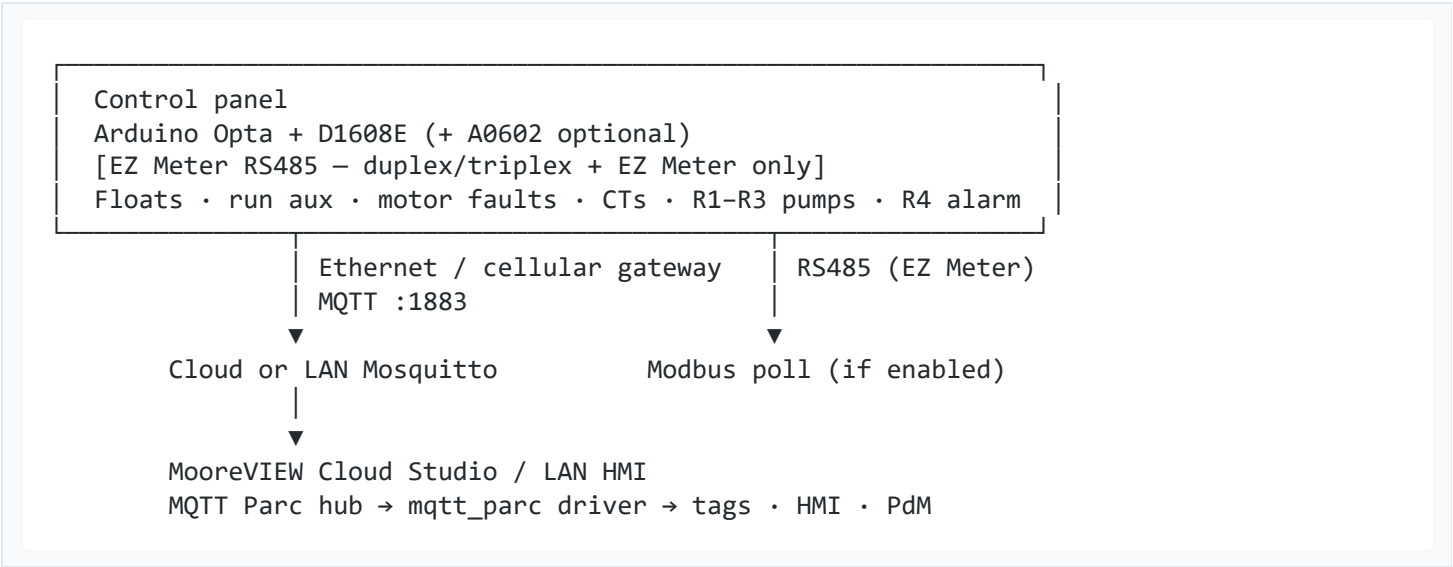
Configuration	Pumps	Floats	ALT	ST program	Cloud template
Simplex	1	4 (start/stop/high/lo)	—	logic/35_lift_simplex.st	lift_station_sin
Duplex	2	4 (HIGH/LEAD/LAG/OFF)	ALT2	logic/36_duplex_lift_station.st	lift_station_dua
Duplex + EZ Meter	2	4	ALT2	logic/38_duplex_lift_station_ezmeter.st	lift_station_dua
Tripdex + EZ Meter	3	5 (+ LAG2)	ALT3	logic/39_triplex_lift_station_ezmeter.st	lift_station_tri

When to use:

Type	Typical site
Simplex	C-store, small tenant, low flow, septic lift
Duplex	Strip mall, school, standard duplex wet well
Duplex + EZ Meter	Main-service PQ monitoring + station alarm on undervolt
Tripdex + EZ Meter	High-flow station, three pumps, facility PQ

Note: Full triplex without EZ Meter is not yet a dedicated ST program — use program 39 or generic `28_alternator_triplex.st` for ALT-only logic.

2. Architecture



See [OPTA_PARC_CLOUD.md](#) and [CSTORE_OPTA_PARC_CLOUD.md](#).

3. Panel bill of materials

Qty	Item	Simplex	Duplex	Triplex
1	Arduino Opta (base only)	✓	—	—
1	Sequent D1608E (slot 1)	—	✓	✓
1	Sequent A0602 (slot 2)	—	Optional	Optional
1	EZ Meter DDS-RGB	—	Optional	Optional
2	0–1 V CT transmitters	2 on I1, I2	2 pumps × 3φ	2 pumps × 3φ (pump 3 CT → expansion)
1–3	Pump contactors	1	2	3
1	Alarm horn/strobe (R4)	✓	✓	✓
4–5	Float switches	4	4	5
1	24 VDC power supply	✓	✓	✓
1	Ethernet / cellular gateway	✓	✓	✓

Full terminal maps: [LIFT-STATION-OPTA-IO-MAP.md](#).

4. Per-configuration build summary

4.1 Simplex (program 35)

Item	Detail
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Hardware	Opta base unit only — no Sequent expansion
Logic	Single pump, start/stop float latch, high/low alarms
Tags	<code>st/fixtures/tags.lift_simplex.json</code>
CTs	2× CT on AI1 / I1 and AI2 / I2 (0–1 V → 0–50 A)
Floats / DIs	I3–I8 digital: START, STOP, HIGH, LO, PUMP_FAIL, run fb
Relays	R1 = MOTOR1_RUN; R4 = station alarm (ALT_FAULT)
I/O binding	Program 35 uses logical tags — map I3 → LVL_START, I4 → LVL_STOP, etc. in Cloud Studio
HMI	mvDraw lift_simplex symbol; no full DUPLEXLS composite
Deploy	Template <code>lift_station_simplex</code> · <code>tagsFromDevice: true</code> · <code>ioBaseOnly: true</code>

4.2 Duplex (program 36)

Item	Detail
Logic	ALT2 alternator, dual HOA faceplates, gen monitoring, fail-to-run
Tags	<code>st/fixtures/tags.duplex_lift_station.json</code>
Floats	X1_I1=HIGH, I2=LEAD, I3=LAG, I4=OFF
Relays	R1/R2 = pumps; R3 spare; R4 = station alarm
Motor faults	Spare DIs on program 36; wire horn to R4 via ALT_FAULT / LVL_HIGH
HMI	@composite/duplexls
Deploy	Template <code>lift_station_dual_duplex</code>

Generate bench project: `npm run generate:duplex-lift-station`

4.3 Duplex + EZ Meter (program 38)

Item	Detail
Logic	Program 36 + facility PQ + R4 station alarm
Tags	<code>st/fixtures/tags.duplex_lift_station_ezmeter.json</code>
Motor faults	X1_I11, X1_I12 → R4
EZ Meter	RS485 on Opta; tags <code>DDS_*</code> , <code>MECH_PQ_*</code>
R4 alarm	<code>LVL_HIGH</code> · <code>MECH_PQ_ALM</code> · motor faults
HMI	@composite/duplexls
Deploy	Template <code>lift_station_dual_duplex_ezmeter</code>

4.4 Triplex + EZ Meter (program 39)

Item	Detail
Logic	ALT3, 5-float ladder, facility PQ, R4 alarm
Tags	st/fixtures/tags.composite_lift_station_ezmeter.json
Floats	X1_I1=HIGH, I2=LEAD, I3=LAG, I4= LAG2 , I5=OFF
Relays	R1–R3 = pumps; R4 = alarm
Motor faults	X1_I13–I15 → R4
Run feedback	X1_I6–I8
Phase fault	X1_I9
Regenerate	node scripts/lift-station/generate-triplex-ezmeter.js after duplex changes
HMI	@composite/triplexls
Deploy	Template lift_station_triplex_ezmeter

5. Shared Opta base unit I/O

Simplex uses the base unit only (§4.1): **A11–A12** = 2 CTs; **I3–I8** = floats/faults; **R1** = pump.

Duplex / triplex add D1608E expansion and use:

Terminal	Tag	Function
R1–R3	MOTOR1_RUN ...	Pump contactors (count per config)
R4	R4	Station alarm (all configurations)
I7, I8	—	Spare digital (I1–I6 = analog CT on duplex/triplex)
A11–A16	Motor CTs	0–1 V → 0–50 A (2 pumps × 3φ)

See I/O map document for complete tables per configuration.

6. Firmware flash (Opta)

- Open `firmware/arduino-opta-mqtt-st/MooreviewOptaMqttSt` in Arduino IDE.
- Tools** → **Board** → **Arduino Opta** (`mbed_opta`).
- Compile flags** by configuration:

Configuration	Flags
Simplex	<i>(none required)</i>
Duplex	-DMV_FIELDBUS=1

Duplex / Triplex + EZ Meter	-DMV_FIELDBUS=1 -DMV_EZMETER=1
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- Libraries: ArduinoJson 7.x, PubSubClient, Arduino_Opta_Blueprint, ArduinoModbus, ArduinoRS485 (EZ Meter builds).
- Upload via USB/Ethernet.

Verify EZ Meter builds: Serial log `[EZM] RS485 EZ Meter poll slave=1`.

7. Opta field configuration (/setup)

Open `http://<opta-ip>/setup`.

Setting	All configs	EZ Meter builds
Expansion slot 1	AFX00005 D1608E	Same
Expansion slot 2	AFX00007 A0602 or Disabled	Same
MQTT broker	Host, port 1883 , user/pass	Same
Global site key	1 (0x0001)	Must match cloud
EZ Meter PQ	—	Nominal 120 V , undervolt 108 V

Test MQTT connection → success + `setup-test` publish. Record **Device ID**.

8. Cloud Studio configuration

8.1 System setup → MQTT Parc

Setting	Value
Enable MQTT Parc hub	✓
Broker URL	mqtt://127.0.0.1:1883 (cloud VM)
Global site key	0x0001
Remote ST execution	✓

8.2 Add Opta driver

Field	Example
Device ID	From Opta /setup
Position ID	bench-duplex-01
Template	See matrix §1

Steps:

- Device **online** in Parc registry
- Sync tags from device**

3. Scan expansions

4. **Program** → **Remote** → **Download & Start** (program from matrix §1)

5. Add HMI composite: **DUPLEXLS** or **TRIPLEXLS** (duplex/triplex); simplex uses mvDraw or custom screen

8.3 EZ Meter PQ (duplex/triplex + EZ Meter only)

Cloud: **System setup** → **MQTT Parc** → **EZ Meter PQ**

Field	Default
Nominal voltage (V)	120
Undervolt threshold (V)	108
Overvolt threshold (V)	132

Saving pushes `MECH_PQ_CFG_*` to connected Opta drivers. Undervolt drives **R4** and interlocks auto pump demand.

Full EZ Meter tag/register reference: see §9 and [EZMETER FACILITY PQ.md](#).

9. EZ Meter (duplex/triplex + EZ Meter builds)

Parameter	Value
Meter	EZ Meter DDS-RGB 2.025, RS485 A/B on Opta
Modbus	9600 8N1, slave 1
Poll	Live ~30 s; energy ~5 min
Key tags	DDS_V_*, DDS_I_*, MECH_PQ_*, MECH_PQ_ALM
R4	ON when MECH_PQ_ALM (undervolt, overvolt, phase loss, imbalance, low PF, freq)

Bench test: With line voltage applied, `DDS_V_A/B/C` $\approx$ 118–122 V. Temporarily set undervolt threshold to 130 V → `MECH_PQ_ALM` and **R4 ON**.

10. Hardware bench test

10.1 All configurations

Step	Pass
24 VDC to Opta + expansion	Expansion detected on <code>/setup</code>
MQTT test	Connected; device online in cloud
ST running	Program → Remote status OK

10.2 Simplex (base unit, 2 CT)

Test	Action	Expected
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Start float	Jumper I3 (+24 V)	LVL_START ON; pump call
Stop float	I4 dry	LVL_STOP; pump off
High alarm	Jumper I5	LVL_HIGH_ALM
Low alarm	Jumper I6 dry	LVL_LO_ALM
Pump run	Auto + call	R1 ON
CT 1	1.0 VDC on I1	AI1 $\approx$ 50 A
CT 2	1.0 VDC on I2	AI2 $\approx$ 50 A

10.3 Duplex / Duplex + EZ Meter

Test	Action	Expected
High float	X1_I1	LVL_HIGH ON
Lead float	X1_I2	LVL_LEAD ON; alternator demand
Phase fault	X1_I7	Pumps interlocked
Motor fault	X1_I11 (EZ Meter build)	R4 ON
Pump 1	Hand START	R1 clicks

10.4 Triplex + EZ Meter

Test	Action	Expected
Lag2 float	X1_I4	3-pump stage demand
Off float	X1_I5 dry	De-stage
Motor 3 fault	X1_I15	R4 ON
Pump 3	Hand START	R3 clicks

10.5 Base CT inputs (all)

Apply **1.0 VDC** to **AI1** → **~50 A** on tag **AI1** .

10.6 EZ Meter (if installed)

DDS_V_* live; undervolt test per §9.

11. Troubleshooting

Symptom	Fix
No expansion DIs	Check 24 V on D1608E; Scan expansions
Float tags inverted	Verify +24 V active when wet

R4 always ON (EZ Meter)	Meter offline → undervolt; check RS485
No DDS_* tags	Reflash with MV_EZMETER=1; check A/B
Simplex tags not updating	Bind logical tags to physical DIs in cloud
Deploy timeout	Enable MQTT Parc hub; check broker auth

12. Quick commissioning checklists

Simplex

#	Task	✓
1	Opta base wired: I1–I2 CTs, I3–I8 floats/faults, R1 pump	
2	Opta flashed; MQTT test OK	
3	Driver added; tags synced	
4	Program 35 deployed	
5	Floats and R1 verified	

Duplex (+ optional EZ Meter)

#	Task	✓
1	Panel wired per I/O map § Duplex	
2	Expansions scanned; AI1–AI6 CTs wired	
3	Flash flags per config matrix	
4	Program 36 or 38 deployed	
5	Floats, pumps, fail-to-run verified	
6	[EZ Meter] RS485 live; R4 alarm tested	

Triplex + EZ Meter

#	Task	✓
1	Panel wired per I/O map § Triplex	
2	5 floats + 3 run fb + 3 motor faults	
3	Program 39 deployed	
4	R1–R3 pumps; R4 alarm verified	
5	EZ Meter PQ tags live	

13. Reference files

Path	Purpose
docs/projects/LIFT-STATION-OPTA-IO-MAP.md	Unified I/O map (this doc's companion)
scripts/lift-station/opta-io-map-data.js	I/O map source data
st/logic/35_lift_simplex.st	Simplex ST
st/logic/36_duplex_lift_station.st	Duplex ST
st/logic/38_duplex_lift_station_ezmeter.st	Duplex + EZ Meter
st/logic/39_triplex_lift_station_ezmeter.st	Triplex + EZ Meter
src/devices/templates/lift_station_*.json	Cloud templates
docs/facilities/EZMETER_FACILITY_PQ.md	EZ Meter Modbus / PQ detail

Build commands:

```
npm run generate:lift-station-io-map-md    # Regenerate I/O map markdown
npm run build:lift-station-io-map-pdf      # Unified I/O map PDF
npm run build:lift-station-parc-cloud-build-pdf  # This guide PDF
npm run generate:duplex-lift-station       # Duplex bench .est project
```